LOTQ-0029 / TIME SERIES / BEGINNER / FIELD GUIDE

Time Series Basics for Price Data

Price series, return conventions, temporal dependence, baselines, and point-in-time pipelines.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0029
CATEGORY	Time Series
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

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How to use this manual

BEGINNER

1. START WITH THE CLOCK

Identify the entity, event time, availability time, session, frequency, and row grain before reading any statistic. Temporal semantics determine what was knowable and whether observations are comparable.

2. CHOOSE THE ECONOMIC OBJECT

Decide whether the question concerns raw prices, adjusted prices, simple returns, log returns, spreads, volume, or forecast errors. A mathematically convenient transform may answer a different trading question.

3. TRACE EVERY WINDOW

For each lag, rolling statistic, label, and resample, draw the exact timestamps used. Mark the decision cutoff and reject any input or fitted parameter that extends beyond it.

4. BENCHMARK BEFORE MODELING

Compare against last-value, zero-return, historical-mean, or seasonal-naive forecasts using the same chronological evaluation. Complexity is justified only by stable incremental decision value.

5. DIAGNOSE THE RESIDUAL PATH

Inspect out-of-sample errors for bias, serial dependence, changing scale, tails, and regime concentration. A good average loss can coexist with unsafe risk calibration.

6. CONNECT RESEARCH TO OPERATION

Record data snapshot, transform state, forecast origin, order time, and fill rule. The backtest is credible only when the same temporal contract can run live.

DECISION STANDARD

A time-series result is ready when its clock, population, transformations, baselines, chronological validation, costs, uncertainty, lineage, monitor, and fallback are explicit and reproducible.

REFERENCE SHEET

Notation and timing contract

BEGINNER

SYMBOL	OBJECT	MEANING
P_t	price level	Selected price field available at time t
r_t	simple return	P_t/P_(t-1)-1 over one declared interval
ell_t	log return	log(P_t/P_(t-1)); additive through time
Delta	sampling gap	Intended or realized elapsed time between rows
L	lookback	Number of prior eligible observations in a window
h	horizon	Number of future periods in an outcome or forecast
gamma_k	autocovariance	Linear co-movement with lag k
rho_k	autocorrelation	Standardized autocovariance at lag k
F_t	information set	All information legally available by decision t
e_t	forecast error	Realized outcome minus archived forecast

TIMING INEQUALITY

For each action require source event time <= availability time <= feature cutoff <= decision time <= order time <= fill time. Outcome windows begin only after the position can exist. This one audit catches many otherwise invisible backtest errors.

MINIMUM SERIES METADATA

Store entity, source, units, timezone, session, frequency, event and availability timestamps, adjustment convention, quality state, and revision. Without these fields a numeric column cannot support a point-in-time claim.

PROCESS MAP

Price-series research workflow

BEGINNER

1. **Define decision clock**
   Horizon, venue, cutoff, latency, and execution

2. **Ingest immutable events**
   Trades, quotes, actions, calendars, and vintages

3. **Build valid series**
   Bars, sessions, quality states, adjustments, and gaps

4. **Create causal transforms**
   Returns, lags, windows, scales, and labels

5. **Diagnose behavior**
   Trend, seasonality, ACF, volatility, and regimes

6. **Run time validation**
   Baselines, purging, walk-forward, costs, uncertainty

7. **Register artifacts**
   Snapshots, state, code, forecasts, orders, and metrics

8. **Monitor and respond**
   Freshness, drift, calibration, fallback, and rollback

HARD GATE

No performance claim is valid until the exact point-in-time series, transform state, chronological split, forecast ledger, and executable fill convention can be replayed from immutable artifacts.

SERIES FOUNDATIONS

A time series is an ordered process

BEGINNER

ONE-SENTENCE MEANING

A time series is a sequence of observations whose order and spacing carry information about how a process evolves.

MEMORY JOGGER

The clock is part of every row, not decorative metadata. Reordering market observations destroys paths, delays, dependence, and the distinction between information available now and outcomes known later.

WHY IT MATTERS IN QUANT RESEARCH

Prices, quotes, returns, spreads, volume, volatility, and model errors are time series with different clocks. Treating them as an unordered table invites invalid sampling, random splits, and features that quietly use future observations.

FORMULA INTUITION

Write a series as X_t, where t identifies an ordered decision time. The joint path $P(X_1,...,X_T)$, not only the marginal distribution of one X_t, determines serial behavior.

SIMPLE MARKET EXAMPLE

Daily closes of 100, 101, 99, and 102 describe a path with gains and losses. The same four values in another order have the same histogram but different returns, drawdown, trend, and trading outcome.

PYTHON / SYSTEM IMPLEMENTATION

Use a monotonic timestamp index, explicit timezone, declared observation grain, and uniqueness checks. Persist both event time and ingestion time so point-in-time reconstruction can distinguish market chronology from pipeline arrival.

CONFUSIONS TO AVOID

A sorted table is not automatically a valid series if timestamps are duplicated, revised, or irregular. Row numbers are not time, and a missing timestamp cannot be inferred safely from neighboring prices.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data contract defines the series key, clock, timezone, grain, and revision policy before any feature is calculated. Every later transform and backtest inherits that temporal contract.

RELATED CONCEPTS AND QUICK RECALL

Related: event time, sampling frequency, lag, information set. Recall task: Explain what information is lost when a price series is shuffled, then name the minimum timestamp fields required for replay.

SERIES FOUNDATIONS

Time index, frequency, and grain

BEGINNER

ONE-SENTENCE MEANING

Frequency states how often observations are intended to occur, while grain states what one row represents and which entity keys make it unique.

MEMORY JOGGER

Ask two questions before analysis: when should a row exist, and what event or interval does that row summarize? A daily close, one-minute bar, and individual trade can share a date but answer different questions.

WHY IT MATTERS IN QUANT RESEARCH

Forecast horizons, transaction costs, seasonality, and effective sample size depend on the observation grid. A strategy tested at the wrong grain may appear smoother, faster, or more executable than the real decision process.

FORMULA INTUITION

For regular data, $t_i=t_0+i\Delta$. For irregular data, retain each realized gap $\Delta_i=t_i-t_{(i-1)}$ because equal row spacing no longer means equal elapsed time.

SIMPLE MARKET EXAMPLE

A Friday close to Monday close is one daily row step but roughly three calendar days. An intraday model that treats this gap like a normal overnight interval misstates both variance and information arrival.

PYTHON / SYSTEM IMPLEMENTATION

Store expected frequency separately from observed timestamps and calculate gap diagnostics by instrument and session. Enforce a composite primary key such as instrument_id plus bar_end_time plus adjustment_version.

CONFUSIONS TO AVOID

Resampling does not create information that never arrived, and filling every expected timestamp can manufacture zero returns. Frequency labels such as daily are incomplete without session, cutoff, and calendar definitions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ingestion layer validates grain and gaps, while the feature store declares the frequency expected by each transform. Model promotion checks that live feeds and research snapshots use the same temporal semantics.

RELATED CONCEPTS AND QUICK RECALL

Related: regular grid, irregular sampling, observation grain, calendar. Recall task: Define the unique key and expected gaps for daily equity bars and for exchange trades.

SERIES FOUNDATIONS

Price levels versus returns

BEGINNER

ONE-SENTENCE MEANING

A price level measures value at a time, while a return measures the change in value over an interval relative to a starting capital base.

MEMORY JOGGER

Prices carry a path and scale; returns express an interval outcome. Most statistical comparisons use returns because raw price levels often drift and differ greatly across instruments.

WHY IT MATTERS IN QUANT RESEARCH

Risk, forecasting, portfolio aggregation, and cross-asset comparison usually require returns rather than levels. Levels remain essential for execution, stops, corporate actions, and long-run equilibrium questions, so the correct representation follows the decision.

FORMULA INTUITION

Simple return $r_t=(P_t-P_{t-1})/P_{t-1}$. Dollar change $\Delta P_t=P_t-P_{t-1}$ answers a different question because it does not scale by starting price.

SIMPLE MARKET EXAMPLE

A move from 20 to 21 and a move from 200 to 201 both gain one dollar, but their simple returns are 5% and 0.5%. A model trained on dollar changes would treat the economic magnitudes as equal.

PYTHON / SYSTEM IMPLEMENTATION

Keep raw and adjusted prices, then derive returns with an explicit field, horizon, and adjustment convention. Unit tests should reproduce hand calculations and reject nonpositive denominators where the formula is undefined.

CONFUSIONS TO AVOID

Returns are not automatically stationary, independent, or normally distributed. Differencing a price changes the target, while using adjusted levels for execution can reference prices that were never tradable.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The canonical price table preserves observable levels; the return service creates versioned analytical outcomes. Strategy, risk, and feature layers consume the representation required by their own metric contracts.

RELATED CONCEPTS AND QUICK RECALL

Related: price difference, adjusted close, stationarity, compounding. Recall task: Contrast dollar change and return for two differently priced assets, then state when raw levels remain necessary.

Simple returns and log returns

BEGINNER

ONE-SENTENCE MEANING

Simple returns measure proportional wealth change, while log returns measure the logarithmic price ratio and add exactly across adjacent time intervals.

MEMORY JOGGER

Simple returns compound by multiplication; log returns aggregate by addition. They are close for small moves but diverge in large moves and have different cross-sectional aggregation properties.

WHY IT MATTERS IN QUANT RESEARCH

Simple returns map directly to portfolio wealth and weighted asset returns over one common interval. Log returns simplify time aggregation, continuous-time models, and some statistical analysis, but cannot be linearly weighted across assets as portfolio returns.

FORMULA INTUITION

r_t=P_t/P_(t-1)-1 and ell_t=log(P_t/P_(t-1))=log(1+r_t). Across time, sum ell_t=log(P_T/P_0), while cumulative simple return=product(1+r_t)-1.

SIMPLE MARKET EXAMPLE

A +50% move followed by -50% gives final wealth 0.75, not one. The log returns sum to log(0.75), correctly preserving the path's compounded loss.

PYTHON / SYSTEM IMPLEMENTATION

Implement both columns from the same validated price pair and label units explicitly. Compare cumulative wealth reconstructed from simple returns with exp of cumulative log returns as a reconciliation test.

CONFUSIONS TO AVOID

Log returns are undefined for nonpositive prices and can obscure a complete loss, whose simple return reaches -100%. Adding simple returns through time or weighting log returns across assets produces incorrect economics.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The transformation registry records return type, price field, horizon, and corporate-action treatment. Downstream metrics declare which return convention they accept, preventing silent formula substitution.

RELATED CONCEPTS AND QUICK RECALL

Related: wealth index, geometric return, arithmetic return, Jensen gap. Recall task: Calculate both return types for a large move and explain why time aggregation and portfolio aggregation use different conventions.

Compounding and the wealth path

BEGINNER

ONE-SENTENCE MEANING

Compounding updates capital by multiplying each period's gross return, making sequence and losses central to long-run wealth.

MEMORY JOGGER

Returns act on the capital that remains, not the original capital every period. A loss requires a larger percentage gain to recover because the recovery starts from a smaller base.

WHY IT MATTERS IN QUANT RESEARCH

Cumulative return, drawdown, strategy comparison, and reinvestment assumptions all depend on a correct wealth path. Arithmetic average return can look attractive while volatility drag produces disappointing compounded growth.

FORMULA INTUITION

W_t=W_0 product_(i=1)^t (1+r_i). Drawdown D_t=W_t/max_(s<=t)W_s-1 and geometric mean g=(W_T/W_0)^(1/T)-1.

SIMPLE MARKET EXAMPLE

A 20% loss reduces 100 to 80, and a 20% gain raises 80 only to 96. Recovering from 80 to 100 requires 25%, showing why symmetric percentage moves are not wealth-symmetric.

PYTHON / SYSTEM IMPLEMENTATION

Use cumulative products with explicit treatment of cash flows, leverage, fees, and missing periods. Reconcile final wealth to the sum of log returns and test bankruptcy or margin constraints before allowing further compounding.

CONFUSIONS TO AVOID

Summing simple returns exaggerates wealth whenever returns vary. A backtest that compounds gross signal returns before deducting costs or ignores capital constraints reports a path the investor could not realize.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The performance engine owns wealth construction and drawdown state rather than leaving compounding to chart code. Risk and allocation services consume the same audited path used in reporting.

RELATED CONCEPTS AND QUICK RECALL

Related: geometric mean, drawdown, volatility drag, path dependence. Recall task: Show the recovery gain after a 40% loss and explain why arithmetic average return does not determine final wealth.

SERIES FOUNDATIONS

Sampling interval changes the signal

BEGINNER

ONE-SENTENCE MEANING

The sampling interval determines which price changes are observed and can materially alter measured dependence, volatility, noise, and apparent predictability.

MEMORY JOGGER

A daily series is not merely an intraday series with fewer rows. Aggregation filters information, mixes events, and changes the ratio of market movement to microstructure noise.

WHY IT MATTERS IN QUANT RESEARCH

Momentum at monthly horizons can coexist with intraday reversal, and realized volatility depends on how often prices are sampled. Model claims must therefore be tied to a horizon rather than presented as timeless properties of an asset.

FORMULA INTUITION

An h-period log return is ell_(t,h)=log(P_t/P_(t-h))=sum_(j=0)^(h-1) ell_(t-j). Under ideal independent increments, variance scales roughly with h, but markets often violate that benchmark.

SIMPLE MARKET EXAMPLE

One-minute midquote returns may show negative serial correlation from bid-ask effects, while daily closes show little correlation. The difference reflects measurement and aggregation, not necessarily a contradiction.

PYTHON / SYSTEM IMPLEMENTATION

Build a horizon matrix that compares return distribution, autocorrelation, volatility, turnover, and cost at several economically meaningful intervals. Compute every horizon from point-in-time data and label whether windows overlap.

CONFUSIONS TO AVOID

Square-root-of-time scaling is an assumption, not a universal law. Resampling to a slower grid can hide jumps or liquidity stress, while faster sampling can amplify stale quotes and bid-ask bounce.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research specification fixes feature, decision, execution, and outcome horizons separately. Validation reports whether conclusions persist across nearby intervals without opportunistically selecting the best one.

RELATED CONCEPTS AND QUICK RECALL

Related: temporal aggregation, realized variance, microstructure noise, horizon. Recall task: Give one mechanism that creates reversal intraday but momentum monthly, and name the diagnostics that separate economics from noise.

OHLCV bars are interval summaries

BEGINNER

ONE-SENTENCE MEANING

An OHLCV bar compresses trades or quotes within an interval into open, high, low, close, and volume fields, discarding the exact within-bar path.

MEMORY JOGGER

A bar is a summary, not a recording of execution order. Identical OHLC values can arise from different paths and therefore imply different stop fills, adverse selection, and attainable trades.

WHY IT MATTERS IN QUANT RESEARCH

Bars support scalable feature computation and medium-frequency research, but they impose assumptions about cutoffs and path. High-low range, close location, and volume can add useful context when their construction is consistent.

FORMULA INTUITION

For interval I_t, O=first valid price, H=max price, L=min price, C=last valid price, and V=sum eligible volume. Each operator requires a source, eligibility rule, and deterministic tie policy.

SIMPLE MARKET EXAMPLE

A bar with open 100, high 105, low 95, close 102 does not reveal whether 95 occurred before 105. A stop-loss and take-profit both touched inside the bar therefore have ambiguous execution ordering.

PYTHON / SYSTEM IMPLEMENTATION

Aggregate from immutable trade or quote events using half-open interval boundaries and exchange calendars. Store event count, first and last event time, source type, and correction version beside the five headline fields.

CONFUSIONS TO AVOID

Vendor bars from different feeds may use different trade-condition filters or session cutoffs. Treating the high and low as simultaneously tradable quotes or assuming a favorable intrabar order creates optimistic backtests.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The bar builder is a versioned data product upstream of features and execution simulation. Strategies that depend on intrabar order must consume finer data or apply conservative ambiguity rules.

RELATED CONCEPTS AND QUICK RECALL

Related: trade conditions, bar boundary, range estimator, intrabar ambiguity. Recall task: Describe two different intrabar paths with the same OHLC values and explain how a conservative simulator should handle them.

Corporate actions and adjusted prices

BEGINNER

ONE-SENTENCE MEANING

Adjusted prices restate historical levels to remove mechanical discontinuities from splits, dividends, and related corporate actions for analytical continuity.

MEMORY JOGGER

A split changes units, not shareholder wealth; a cash dividend transfers value from the company to the holder. The adjustment method must match whether the analysis targets price return or total return.

WHY IT MATTERS IN QUANT RESEARCH

Unadjusted splits create false jumps that contaminate returns, volatility, outliers, and model training. Total-return research also needs dividend treatment, while execution simulations require the actual historical prices and share quantities.

FORMULA INTUITION

For a split factor a_t, adjusted historical price often rescales prior prices by a cumulative factor. Total return uses $(P_t + \text{cash distribution during interval})/P_{(t-1)}-1$ under an explicit reinvestment convention.

SIMPLE MARKET EXAMPLE

A 2-for-1 split can make a 100 close become 50 without an economic loss. An unadjusted return reports -50%, while a correctly adjusted series reports only the genuine market move.

PYTHON / SYSTEM IMPLEMENTATION

Keep raw prices, action records, adjustment factors, ex-dates, pay dates, and vendor vintages separately. Rebuild adjusted series deterministically and reconcile split continuity and dividend wealth against independent totals.

CONFUSIONS TO AVOID

Adjusted prices are synthetic and should not be used as historical limit-order prices. Backward-adjusted datasets can revise past values when new actions arrive, so cached features need lineage and recomputation rules.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The security-master and corporate-action services produce versioned adjustment artifacts. Research chooses price-return or total-return semantics explicitly, and execution retrieves raw point-in-time prices.

RELATED CONCEPTS AND QUICK RECALL

Related: split factor, dividend, total return, point-in-time data. Recall task: Explain which data are used for a factor backtest versus an execution simulation around a stock split.

Bid, ask, mid, and last trade

BEGINNER

ONE-SENTENCE MEANING

Bid, ask, midpoint, and last trade are different price observations that represent executable sides, a quote summary, and a completed transaction respectively.

MEMORY JOGGER

There is no single true intraday price field. The correct choice depends on whether the task is valuing inventory, measuring movement, or estimating what an order could execute.

WHY IT MATTERS IN QUANT RESEARCH

Using last trades can introduce bounce and staleness, while midquotes can understate execution cost. Signal returns, mark-to-market PnL, and fill simulation should use fields consistent with the decision and side of trade.

FORMULA INTUITION

$Mid_t=(Bid_t+Ask_t)/2$ and $spread_t=Ask_t-Bid_t$. A market buy faces the ask plus impact; a market sell faces the bid minus impact under a simplified book model.

SIMPLE MARKET EXAMPLE

If quotes remain 99.95 by 100.05 while trades alternate at bid and ask, last-trade returns oscillate despite an unchanged midpoint. The apparent reversal is microstructure bounce rather than changing fair value.

PYTHON / SYSTEM IMPLEMENTATION

Store quote and trade streams with source timestamps, conditions, sizes, and sequence identifiers. Derive midpoint returns and effective spread, then reconcile each simulated fill to the contemporaneous book state.

CONFUSIONS TO AVOID

Midpoint is not guaranteed executable, especially for large or latency-sensitive orders. Last prices from thin instruments may be stale, and crossed or locked markets require explicit quality handling.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The market-data layer exposes distinct semantic fields, and the execution model chooses a side-aware benchmark. Feature code cannot silently substitute close, midpoint, or last trade based on availability.

RELATED CONCEPTS AND QUICK RECALL

Related: quoted spread, effective spread, bid-ask bounce, mark price. Recall task: Construct a flat-midpoint sequence with alternating last trades and explain the false autocorrelation it creates.

MARKET DATA CONSTRUCTION

Event time versus clock time

BEGINNER

ONE-SENTENCE MEANING

Clock time samples fixed elapsed intervals, while event time advances when trades, quotes, volume, or another market event occurs.

MEMORY JOGGER

Clock bars compare equal durations; event bars compare roughly equal activity. Quiet periods stretch event time, and active periods compress many market changes into a short wall-clock span.

WHY IT MATTERS IN QUANT RESEARCH

Event-time sampling can reduce heterogeneity in information per row and help model high-frequency dynamics. Clock-time sampling is easier for portfolios, calendars, and latency budgets, so neither grid is universally superior.

FORMULA INTUITION

A volume bar closes when cumulative eligible volume since the prior boundary reaches q. A clock bar closes at predetermined t_k, regardless of whether one event or ten thousand events occurred.

SIMPLE MARKET EXAMPLE

A 10,000-share bar may take twenty minutes at lunch but five seconds after a major announcement. Its return spans unequal elapsed time even though activity is more comparable.

PYTHON / SYSTEM IMPLEMENTATION

Preserve the raw event stream, then create deterministic bar assignments with boundary IDs and partial-bar rules. Store elapsed duration, event count, and clock timestamps on every event-time observation.

CONFUSIONS TO AVOID

Event bars do not eliminate dependence or selection bias, and threshold crossing can create variable overshoot. Comparing event-time returns without duration metadata can confuse activity normalization with forecasting skill.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The sampling service publishes multiple approved grids from one immutable event source. Research declares which clock governs features, predictions, orders, and evaluation before model fitting.

RELATED CONCEPTS AND QUICK RECALL

Related: tick bars, volume bars, imbalance bars, duration. Recall task: Compare the meaning of one observation in a five-minute bar and a fixed-volume bar during calm and active markets.

MARKET DATA CONSTRUCTION

Calendars, sessions, and time zones

BEGINNER

ONE-SENTENCE MEANING

Trading calendars and time zones define which timestamps belong to a session, how days are labeled, and where holidays, breaks, and daylight-saving changes occur.

MEMORY JOGGER

A timestamp without a timezone is incomplete. A date without a market calendar can merge different sessions or compare observations that were never simultaneously available.

WHY IT MATTERS IN QUANT RESEARCH

Overnight returns, daily closes, global asset alignment, and feature publication depend on session boundaries. Errors around daylight saving or holidays can create duplicated bars, gaps, and accidental lookahead across markets.

FORMULA INTUITION

Convert storage timestamps to a canonical timezone such as UTC, but derive session labels from the venue's local calendar. Session return may decompose into close-to-open plus open-to-close components with distinct information sets.

SIMPLE MARKET EXAMPLE

At 4:00 p.m. New York, a European cash market is already closed and an Asian session is from a different calendar day. Joining all rows on a naive date can feed unavailable European information into an earlier decision.

PYTHON / SYSTEM IMPLEMENTATION

Use exchange-maintained calendars with scheduled opens, closes, breaks, half-days, and timezone-aware timestamps. Test known daylight-saving transitions and version the calendar because venues can announce exceptions.

CONFUSIONS TO AVOID

Converting to UTC does not remove the need for local session semantics. Forward-filling a closed market through another market's open manufactures synchronous data and understates stale-price risk.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar service assigns session_id and expected interval boundaries before aggregation. Cross-market features use explicit availability cutoffs rather than naive calendar-date joins.

RELATED CONCEPTS AND QUICK RECALL

Related: session label, holiday, daylight saving, asynchronous market. Recall task: Explain how a date-only join leaks information across regions and specify a safe as-of join rule.

Missing bars and stale prices

BEGINNER

ONE-SENTENCE MEANING

A missing observation means no valid value was recorded, while a stale price repeats an old value despite the passage of time or market activity.

MEMORY JOGGER

No row, no trade, and no price change are three different states. Encoding all three as zero return erases the mechanism that produced the absence.

WHY IT MATTERS IN QUANT RESEARCH

Missingness can signal halts, illiquidity, feed failure, or calendar closure, each with different research implications. Staleness creates artificial low volatility and delayed jumps when prices eventually update.

FORMULA INTUITION

Age_t=t-last_valid_update_time and stale_t=1[Age_t>threshold]. Coverage should be measured against expected eligible observations, not against whatever rows happen to exist.

SIMPLE MARKET EXAMPLE

A thin bond prints at 100 on Monday and not again until Thursday. Filling every day at 100 creates three zero returns, then assigns the full move to Thursday and distorts dependence with liquid assets.

PYTHON / SYSTEM IMPLEMENTATION

Attach reason codes, last-update age, venue status, and observation quality to each interval. Use state-aware missing policies, report coverage by asset and period, and quarantine feed outages before estimation.

CONFUSIONS TO AVOID

Forward fill is an assumption that value remained unchanged, not a neutral cleaning step. Dropping missing rows can select the most liquid states and bias costs, volatility, and cross-sectional comparisons.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quality layer distinguishes closures, halts, no-activity intervals, and technical failures. Feature and risk services consume quality masks and apply policies approved for each data type.

RELATED CONCEPTS AND QUICK RECALL

Related: missingness mechanism, halt, liquidity, observation age. Recall task: Contrast how a closed venue, trading halt, and feed outage should appear in the dataset and downstream analysis.

Bad ticks and genuine jumps

BEGINNER

ONE-SENTENCE MEANING

Bad ticks are erroneous observations, while genuine jumps are real discontinuous market moves that may carry the most important risk information in the sample.

MEMORY JOGGER

First establish provenance, then decide treatment. An extreme value is not evidence of an error merely because it changes the model or makes a chart uncomfortable.

WHY IT MATTERS IN QUANT RESEARCH

One corrupted price can dominate returns, volatility, correlations, and labels, but deleting authentic crises sanitizes the very tails a trading system must survive. Data correction and robust estimation are separate decisions.

FORMULA INTUITION

A local return score z_t=(r_t-median window)/robust_scale can flag review candidates, not prove error. Cross-venue, quote, trade-condition, and corporate-action checks provide stronger evidence.

SIMPLE MARKET EXAMPLE

A price of 1.02 between trades near 102 is likely a decimal error; a verified 15% earnings gap across venues is genuine. Both are extreme, but only the first should be corrected or quarantined.

PYTHON / SYSTEM IMPLEMENTATION

Preserve raw values, quality flags, evidence, correction author, and replacement lineage. Run deterministic cross-source and continuity checks, then publish both corrected and raw audit views.

CONFUSIONS TO AVOID

Winsorizing before diagnosis can hide pipeline failures, while thresholding on a full-sample distribution leaks future scale. Genuine jumps should remain in risk and robustness analysis even if a model uses robust transforms.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quality gate resolves suspect observations before feature computation and exposes unresolved states. Statistical pipelines can downweight valid extremes only under a declared estimator policy, never as silent data repair.

RELATED CONCEPTS AND QUICK RECALL

Related: outlier, jump, robust scale, data lineage. Recall task: List the evidence needed to distinguish a decimal error from a real gap and state which artifacts must remain auditable.

Resampling and bar aggregation

BEGINNER

ONE-SENTENCE MEANING

Resampling converts a series to a new time grid by aggregating finer observations or assigning values to coarser interval boundaries.

MEMORY JOGGER

Every resample embeds rules about boundary inclusion, timezone, source fields, and empty intervals. Those rules can move returns across periods and change feature timing.

WHY IT MATTERS IN QUANT RESEARCH

Consistent bars are necessary for comparable signals and portfolio synchronization. Incorrect close labels or boundary choices can create a one-bar lookahead even when the underlying events are timestamped correctly.

FORMULA INTUITION

For half-open intervals (a,b], close uses the last eligible event at or before b and after a. Return at b must use only bar values whose availability is no later than the decision cutoff.

SIMPLE MARKET EXAMPLE

A trade exactly at 10:00:00 can belong to the 9:59 or 10:00 bar depending on convention. If research and production disagree, identical event data produce different signals and fills.

PYTHON / SYSTEM IMPLEMENTATION

Centralize resampling with explicit origin, offset, interval closure, label side, calendar, and empty-bin policy. Add golden tests for boundary events and reconcile aggregated volume to the source stream.

CONFUSIONS TO AVOID

Downsampling loses path detail and cannot support optimistic intrabar execution logic. Upsampling with filled values creates rows but not new observations, so uncertainty and sample size must not count them as fresh evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The bar service owns grid conversion and versions its configuration with each dataset. Feature jobs reference a bar artifact ID rather than invoking local resample defaults.

RELATED CONCEPTS AND QUICK RECALL

Related: downsampling, upsampling, half-open interval, boundary label. Recall task: Place events exactly on two adjacent boundaries and demonstrate how closure and labeling choices affect a return feature.

Alignment and as-of joins

BEGINNER

ONE-SENTENCE MEANING

Time-series alignment determines which observations from different sources are considered contemporaneously available, often using the latest valid record at or before a decision time.

MEMORY JOGGER

Equal timestamps do not guarantee equal knowledge, and equal dates can be dangerously misleading. Join on availability semantics, not convenience or nearest absolute distance.

WHY IT MATTERS IN QUANT RESEARCH

Multi-asset signals, fundamentals, macro data, and benchmarks arrive on different schedules. A correct as-of join prevents future releases or later market closes from entering an earlier prediction.

FORMULA INTUITION

For decision time t, select source record i with availability_i=max{availability_j: availability_j<=t}, subject to a freshness limit. Event date alone is insufficient when publication occurs later.

SIMPLE MARKET EXAMPLE

A quarterly value labeled March 31 but released May 5 cannot be used in an April 15 signal. A backward as-of join on release availability correctly leaves the prior filing in place.

PYTHON / SYSTEM IMPLEMENTATION

Store event, publication, vendor, ingestion, and revision timestamps, then implement backward joins with tolerance and provenance columns. Unit tests should insert a future record and verify that all earlier features remain unchanged.

CONFUSIONS TO AVOID

Nearest joins can select the future, and unlimited backward joins can use dangerously stale records. End-of-day data from one region may be unavailable before another region's same-date close.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The point-in-time feature store performs approved temporal joins and records the source row for every derived value. Backtests reject features lacking availability lineage or freshness status.

RELATED CONCEPTS AND QUICK RECALL

Related: availability time, publication lag, freshness, point-in-time join. Recall task: Write the join rule for a macro release and a cross-market close, including direction and maximum staleness.

Differencing removes level changes

BEGINNER

ONE-SENTENCE MEANING

Differencing replaces a series with changes between observations, often reducing trend and shifting the question from level to movement.

MEMORY JOGGER

Subtract the past from the present to study increments. The operation can stabilize a drifting series, but it also discards long-run level information and introduces dependence when overused.

WHY IT MATTERS IN QUANT RESEARCH

Returns are scaled differences of prices, and differences of rates or spreads can be more suitable for short-horizon models than their persistent levels. Transformation choice must reflect an economic target, not only a statistical test.

FORMULA INTUITION

First difference Delta X_t=X_t-X_(t-1); second difference Delta^2 X_t=Delta X_t-Delta X_(t-1). Log difference of a positive price equals its log return.

SIMPLE MARKET EXAMPLE

A price level with steady upward drift can have a stable distribution of daily changes. Differencing a truly stable spread, however, may create unnecessary negative autocorrelation and obscure mean reversion in the level.

PYTHON / SYSTEM IMPLEMENTATION

Create differenced fields with explicit lag, grouping key, calendar, and gap handling. Preserve the source level and mark discontinuities so a multi-day gap is not mistaken for a one-period change.

CONFUSIONS TO AVOID

Differencing is not automatic proof of stationarity and cannot repair structural breaks or bad data. Repeated differencing can amplify measurement noise and make forecasts difficult to translate back to economic levels.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The transform library versions difference order and lag and fits only where required. Model reports compare level, difference, and return formulations using the same time-safe splits.

RELATED CONCEPTS AND QUICK RECALL

Related: unit root, return, integration order, noise amplification. Recall task: Give a case where differencing helps and one where it damages the useful signal, then state the diagnostics for each.

Lags, leads, and information direction

BEGINNER

ONE-SENTENCE MEANING

A lag references an earlier observation, while a lead references a later observation and is normally valid only as a label or retrospective diagnostic.

MEMORY JOGGER

Features point backward from the decision; outcomes point forward. A sign or shift error can turn tomorrow's return into today's predictor without obvious code failure.

WHY IT MATTERS IN QUANT RESEARCH

Lagged values define autoregressive features, delayed market relationships, and state variables. Leads define prediction targets, but they must remain isolated from preprocessing and feature selection.

FORMULA INTUITION

Lag k: $L^k X_t=X_{t-k}$. An h-step forward return label $y_t=P_{t+h}/P_t-1$ uses future prices and therefore belongs only on the outcome side.

SIMPLE MARKET EXAMPLE

Yesterday's close-to-close return may be known before today's decision, while tomorrow's return is the target. A negative shift in a dataframe library can silently place the latter in the feature matrix.

PYTHON / SYSTEM IMPLEMENTATION

Name columns with direction and horizon, enforce positive lag values in feature APIs, and add cutoff tests that mutate future rows. Maintain separate feature and label tables joined only after both are created.

CONFUSIONS TO AVOID

A lagged daily close may still be unavailable at an intraday decision if the session labeling is wrong. Shifting rows is not equivalent to shifting elapsed time when observations are irregular or missing.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature compiler permits backward references through a controlled lag operator, while the label service owns forward windows. Lineage audits verify that feature availability never exceeds the prediction timestamp.

RELATED CONCEPTS AND QUICK RECALL

Related: shift operator, forecast horizon, label, lookahead bias. Recall task: Draw the timestamps used by a two-day lag and a five-day forward label, then identify the earliest legal prediction time.

Rolling windows are local estimators

BEGINNER

ONE-SENTENCE MEANING

A rolling statistic recomputes a summary from a moving window, trading responsiveness to recent conditions against estimation noise and dependence between adjacent estimates.

MEMORY JOGGER

The window forgets old observations one at a time and admits new ones one at a time. Neighboring outputs share most data, so a smooth plot does not represent many independent estimates.

WHY IT MATTERS IN QUANT RESEARCH

Rolling means, volatilities, correlations, highs, and ranks describe local market state and drive adaptive risk rules. Window length, minimum observations, and timestamp alignment materially determine both behavior and leakage risk.

FORMULA INTUITION

For length L, rolling mean $m_t=(1/L) \sum_{i=0}^{L-1} X_{t-i}$. A feature used at t must exclude X_t if X_t is not finalized before the decision.

SIMPLE MARKET EXAMPLE

A 20-day volatility estimate at today's open cannot use today's close. A one-day alignment mistake makes the estimate react perfectly to a return that had not yet occurred.

PYTHON / SYSTEM IMPLEMENTATION

Implement rolling operators with declared closed side, min_periods, lag, and group key. Store window start, end, eligible count, and missing count so each value is reproducible.

CONFUSIONS TO AVOID

A longer window is not simply better because it can mix obsolete regimes, while a shorter window can be dominated by noise. Overlapping windows invalidate naive significance tests on the sequence of estimates.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature store materializes lagged rolling states with lineage. Monitoring compares live window coverage and distribution with training and alerts when missingness or regime speed invalidates the chosen length.

RELATED CONCEPTS AND QUICK RECALL

Related: moving average, rolling volatility, overlap, window length. Recall task: Specify a leakage-safe 20-day volatility feature available at the open and list the metadata required to reproduce one value.

TRANSFORMS AND DESCRIPTORS

Expanding windows accumulate history

BEGINNER

ONE-SENTENCE MEANING

An expanding statistic uses all eligible observations from a fixed start through the current cutoff, increasing stability while gradually reducing adaptation to recent change.

MEMORY JOGGER

The sample grows but never forgets. Early values are fragile, and late values can be anchored by regimes that no longer describe current markets.

WHY IT MATTERS IN QUANT RESEARCH

Expanding estimates are useful for long-run baselines, online normalization, and evidence tracking when structural stability is plausible. They provide a clear no-future-data path but can respond too slowly to breaks.

FORMULA INTUITION

Expanding mean m_t=(1/n_t) sum_(i<=t) X_i over eligible records. Recursive update m_t=m_(t-1)+(X_t-m_(t-1))/n_t avoids recomputing the full history.

SIMPLE MARKET EXAMPLE

An expanding average daily volume becomes stable after years of data but may understate the importance of a recent permanent liquidity increase. A rolling baseline adapts faster but fluctuates more.

PYTHON / SYSTEM IMPLEMENTATION

Declare the anchor date, warm-up count, eligibility policy, and reset rules. Use numerically stable online algorithms and store state checkpoints so production values can be replayed exactly.

CONFUSIONS TO AVOID

More history can reduce variance while increasing bias to the current regime. Restarting an expanding window after observing a break is a model decision that requires validation, not a harmless cleaning step.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The stateful feature service maintains expanding sufficient statistics and checkpoint versions. Model governance defines when a new population justifies a reset or segmented baseline.

RELATED CONCEPTS AND QUICK RECALL

Related: online estimator, recursive update, warm-up, structural break. Recall task: Compare an expanding and rolling volume baseline after a permanent market change, including the bias-variance tradeoff.

Moving averages and smoothing

BEGINNER

ONE-SENTENCE MEANING

Moving averages smooth short-term variation to estimate a local level, with each weighting scheme defining how quickly old information loses influence.

MEMORY JOGGER

Smoothing suppresses noise by accepting delay. The smoother the line, the more slowly it usually recognizes a genuine turning point.

WHY IT MATTERS IN QUANT RESEARCH

Simple and exponential moving averages support trend features, baseline forecasts, volatility estimates, and monitoring. Their value comes from a defined decision horizon, not from visual appeal on a historical chart.

FORMULA INTUITION

SMA_t=(1/L)sum_(i=0)^(L-1)X_(t-i). EWMA_t=alpha X_t+(1-alpha)EWMA_(t-1), with effective memory controlled by alpha.

SIMPLE MARKET EXAMPLE

A fast EWMA reacts quickly to a price jump but retains more noise, while a slow EWMA changes gradually and can remain stale through a reversal. A crossover is therefore a comparison of two lag structures.

PYTHON / SYSTEM IMPLEMENTATION

Fit or choose smoothing parameters inside time-series validation and lag the result to the legal decision time. Initialize recursively with a documented seed and test equivalence between batch and streaming calculations.

CONFUSIONS TO AVOID

A smooth backtest input is not proof of predictability, and centered moving averages use future values. Optimizing many lengths on one history creates selection bias even when each rule is simple.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The transform service exposes approved causal filters and stores parameter, initialization, and cutoff. Strategy evaluation measures forecast value, turnover, and delay rather than rewarding visual smoothness.

RELATED CONCEPTS AND QUICK RECALL

Related: low-pass filter, EWMA, latency, crossover. Recall task: Explain the reaction-delay tradeoff between two smoothing constants and identify why a centered average is illegal live.

Normalization must be point in time

BEGINNER

ONE-SENTENCE MEANING

Time-series normalization expresses values relative to a scale estimated from past data, making magnitudes comparable while creating another learned, time-varying state.

MEMORY JOGGER

A z-score is only as honest as the mean and scale behind it. Full-history normalization lets future calm or crisis periods reshape every earlier feature.

WHY IT MATTERS IN QUANT RESEARCH

Volatility scaling, rolling z-scores, and robust standardization help compare assets and stabilize models. They also amplify errors when scale is tiny and can hide economically important changes in absolute risk.

FORMULA INTUITION

$z_t=(X_t-m_{t-1})/s_{t-1}$ for a feature available at t, with center and scale fit only through the prior legal cutoff. Robust variants use median and MAD.

SIMPLE MARKET EXAMPLE

A spread of two points is extreme when its trailing scale is 0.5 but ordinary when scale is four. Computing scale from the full sample can make a historical crisis appear less unusual than it was at the time.

PYTHON / SYSTEM IMPLEMENTATION

Create a pipeline object that fits center, scale, clipping, and fallback on training data, then transforms later periods unchanged or via declared online updates. Store denominators and zero-scale flags.

CONFUSIONS TO AVOID

Normalization does not remove nonstationarity; it can merely express it differently. Cross-sectional and time-series z-scores answer different questions, and clipping after looking at outcomes consumes validation evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature pipeline registers every learned normalization state and applies it consistently in research and production. Monitoring reports raw values alongside normalized scores to preserve economic context.

RELATED CONCEPTS AND QUICK RECALL

Related: z-score, robust scaling, volatility targeting, preprocessing leakage. Recall task: Design a trailing z-score available before the close and list the safeguards for small denominators and regime changes.

DEPENDENCE AND BEHAVIOR

Stationarity is a modeling contract

BEGINNER

ONE-SENTENCE MEANING

Stationarity means selected distributional properties remain stable under time shifts, providing a reference process from which historical behavior can inform future behavior.

MEMORY JOGGER

A stationary model assumes that when the clock origin changes, the quantities it uses do not change in a material way. The assumption concerns a target and horizon, not a permanent label attached to an asset.

WHY IT MATTERS IN QUANT RESEARCH

Means, variances, autocorrelations, and forecast errors are easier to estimate when their governing process is stable. Financial series often achieve only local or conditional stability, so windows and regimes must be explicit.

FORMULA INTUITION

Weak stationarity requires constant $E[X_t]$, finite constant $Var(X_t)$, and $Cov(X_t,X_{t-k})$ depending on lag k rather than calendar time. Strict stationarity makes the full joint distribution shift-invariant.

SIMPLE MARKET EXAMPLE

Daily returns may have a roughly stable mean over a period while volatility changes dramatically, violating constant variance. A model can still work conditionally if volatility state is modeled and validated.

PYTHON / SYSTEM IMPLEMENTATION

Plot rolling location, scale, quantiles, and dependence; compare subperiods and apply tests as supporting evidence rather than verdicts. Define the population and reset or fallback policy before live use.

CONFUSIONS TO AVOID

Failure to reject a unit-root or stationarity test is not proof of the desired process, especially in short dependent samples. Transforming to returns does not guarantee stable variance, tails, or relationships.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model specification states which moments are assumed stable, over what window, and under which state controls. Monitoring tests those exact properties and links failures to a retraining or risk response.

RELATED CONCEPTS AND QUICK RECALL

Related: weak stationarity, unit root, local stationarity, distribution drift. Recall task: Name which moments of daily returns could be stable while volatility is not, and explain what the model must condition on.

DEPENDENCE AND BEHAVIOR

Trend and deterministic time structure

BEGINNER

ONE-SENTENCE MEANING

A trend is a systematic change in the level or expected value of a series over time, whether represented deterministically or generated by persistent stochastic shocks.

MEMORY JOGGER

A rising line can reflect a deterministic clock effect, accumulated random innovations, inflation, or compounding. Removing a fitted line is justified only when that representation matches the decision.

WHY IT MATTERS IN QUANT RESEARCH

Trending levels create spurious regressions and unstable baselines, while trend-following strategies intentionally seek persistence in returns or price direction. Distinguishing level trend from forecastable return is essential.

FORMULA INTUITION

A deterministic model may write $X_t=a+bt+\epsilon_t$. A stochastic trend such as random walk $X_t=X_{t-1}+\epsilon_t$ accumulates shocks and does not revert to one fixed line.

SIMPLE MARKET EXAMPLE

Two unrelated asset prices can both rise over a decade and produce high level correlation. Their returns may be nearly unrelated, showing that shared clock direction was mistaken for a tradable relationship.

PYTHON / SYSTEM IMPLEMENTATION

Compare levels, differences, returns, detrended residuals, and out-of-sample forecasts. Fit trend terms inside each training window and store the origin, units, and extrapolation rule.

CONFUSIONS TO AVOID

Detrending with the full history leaks future slope and break information. A statistically fitted trend is not evidence of economic momentum, and polynomial trends can extrapolate catastrophically.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The modeling layer requires a declared trend mechanism and benchmark. Feature review rejects level relationships that disappear under time baselines unless a validated long-run mechanism supports them.

RELATED CONCEPTS AND QUICK RECALL

Related: random walk, drift, detrending, spurious regression. Recall task: Contrast deterministic and stochastic trends and describe how each should be evaluated out of sample.

Seasonality and calendar effects

BEGINNER

ONE-SENTENCE MEANING

Seasonality is a recurring pattern tied to a known position in a cycle such as hour, weekday, month, auction, or contract schedule.

MEMORY JOGGER

The same clock location may carry similar flows, liquidity, or risk. A valid seasonal effect repeats prospectively and survives changes in calendar composition and trading costs.

WHY IT MATTERS IN QUANT RESEARCH

Intraday volume curves, overnight gaps, month-end flows, and funding cycles affect feature baselines and execution. Accounting for them can prevent a model from confusing normal calendar structure with anomaly or alpha.

FORMULA INTUITION

X_t=level_t+s_(calendar(t))+epsilon_t is a basic decomposition. Seasonal estimates require enough independent cycles, not merely many rows within a few repeated days.

SIMPLE MARKET EXAMPLE

Volume is often high near the open and close and low midday. Comparing a noon observation with a full-day average falsely labels normal activity as unusually weak.

PYTHON / SYSTEM IMPLEMENTATION

Create calendar features from information known in advance and estimate seasonal profiles only on training cycles. Normalize within comparable session buckets and track holidays, half-days, and contract rolls separately.

CONFUSIONS TO AVOID

Searching many calendar slices produces false discoveries, and daylight-saving or venue changes can shift patterns. Seasonality in volume does not automatically imply seasonality in profitable returns after costs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar feature service publishes stable session-position keys, while monitoring compares each bucket with its own historical baseline. Strategy validation reserves later cycles to test persistence.

RELATED CONCEPTS AND QUICK RECALL

Related: intraday curve, day-of-week effect, seasonal adjustment, multiple testing. Recall task: Design a test for an opening-volume pattern that counts independent days and handles half-day sessions.

Autocorrelation measures linear memory

BEGINNER

ONE-SENTENCE MEANING

Autocorrelation measures the linear relationship between a series and a lagged version of itself at a specified lag.

MEMORY JOGGER

Ask whether high or low values tend to be followed by similarly signed deviations after k steps. The answer depends on transformation, horizon, sampling grid, and market state.

WHY IT MATTERS IN QUANT RESEARCH

Autocorrelation informs momentum, mean reversion, standard errors, effective sample size, and baseline model choice. Small return autocorrelation can coexist with strong autocorrelation in volatility, volume, spreads, or positions.

FORMULA INTUITION

rho_k=Cov(X_t,X_(t-k))/Var(X_t) under a stable variance convention. Sample estimates use paired observations and become noisier at longer lags because fewer pairs remain.

SIMPLE MARKET EXAMPLE

Daily returns may have rho_1 near zero while absolute returns have persistent positive autocorrelation. The series may be unpredictable in direction but predictable in risk magnitude.

PYTHON / SYSTEM IMPLEMENTATION

Compute ACF with sample counts, confidence guidance, and subperiod overlays; inspect raw, absolute, and squared returns. Use dependence-aware uncertainty and avoid pooling incompatible regimes.

CONFUSIONS TO AVOID

Autocorrelation is not causal and can arise from stale prices, overlapping windows, bid-ask bounce, or smoothing. A nonzero sample spike among many lags may be chance selection.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The diagnostics service evaluates lag structure before model fitting and sets valid resampling blocks and standard errors. Monitoring compares the same lag set used in research rather than choosing new lags after drift appears.

RELATED CONCEPTS AND QUICK RECALL

Related: serial correlation, ACF, effective sample size, momentum. Recall task: Explain why returns and absolute returns can show different memory, then name two artificial sources of lag-one correlation.

Partial autocorrelation isolates a lag

BEGINNER

ONE-SENTENCE MEANING

Partial autocorrelation measures the incremental linear association at lag k after accounting for the shorter intervening lags.

MEMORY JOGGER

Ordinary autocorrelation includes direct and chained relationships. Partial autocorrelation asks whether lag k adds information beyond lags 1 through k-1 in a linear projection.

WHY IT MATTERS IN QUANT RESEARCH

PACF helps describe autoregressive order and separates a direct lag from persistence propagated through nearer observations. It is a diagnostic, not a complete model-selection rule for nonlinear or changing markets.

FORMULA INTUITION

PACF(k) is the coefficient on $X_{(t-k)}$ in a regression of X_t on $X_{(t-1)},\dots,X_{(t-k)}$, under a consistent normalization. Estimation uncertainty grows with k.

SIMPLE MARKET EXAMPLE

An AR(1) process can have autocorrelation that decays across many lags but PACF concentrated at lag one. The farther ACF spikes reflect propagation rather than separate direct mechanisms.

PYTHON / SYSTEM IMPLEMENTATION

Calculate ACF and PACF on a training window with a fixed maximum lag and enough observations per parameter. Validate any chosen autoregressive order through rolling forecasts, residual checks, and cost-aware decisions.

CONFUSIONS TO AVOID

A clean PACF cutoff is a textbook pattern, not a guarantee in finite noisy data. Seasonality, missing values, regime mixtures, and estimated trends can create misleading spikes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The baseline-model workflow uses PACF to propose small candidate orders, then time-series validation chooses among them. Production monitoring checks residual dependence rather than repeatedly retuning order from one chart.

RELATED CONCEPTS AND QUICK RECALL

Related: AR model, ACF, lag order, residual diagnostics. Recall task: Describe the ACF and PACF pattern of a simple AR(1) process and explain why validation remains necessary.

White noise is a benchmark, not a synonym for random

BEGINNER

ONE-SENTENCE MEANING

White noise is a zero-mean uncorrelated sequence with constant finite variance, though it need not be independent or normally distributed.

MEMORY JOGGER

White means no linear color across time. Nonlinear dependence, changing tails, or conditional variance can remain even when every linear autocorrelation is zero.

WHY IT MATTERS IN QUANT RESEARCH

Residuals should often approximate white noise after a useful linear model, because remaining serial structure indicates missed forecast information. Returns can be directionally white while their squared values remain highly predictable.

FORMULA INTUITION

For white noise epsilon_t, E[epsilon_t]=0, Var(epsilon_t)=sigma^2, and Cov(epsilon_t,epsilon_(t-k))=0 for k not equal to zero. IID Gaussian noise adds stronger distributional assumptions.

SIMPLE MARKET EXAMPLE

A sequence epsilon_t=z_t z_(t-1) with independent zero-mean z values can be uncorrelated yet dependent. Linear ACF alone therefore cannot establish independence.

PYTHON / SYSTEM IMPLEMENTATION

Inspect residual ACF, squared and absolute residual ACF, distribution stability, and runs or nonlinear tests. Use portmanteau tests with model degrees of freedom and predeclared lag ranges.

CONFUSIONS TO AVOID

A nonsignificant autocorrelation does not prove white noise, and white noise does not imply safe tails. Normality, independence, and constant conditional variance are separate claims.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model-evaluation service treats whiteness as one residual quality gate alongside calibration, tail behavior, and stability. Failure routes the model back to specification review rather than automatic order expansion.

RELATED CONCEPTS AND QUICK RECALL

Related: innovation, residual, independence, heteroskedasticity. Recall task: State the assumptions that distinguish uncorrelated white noise from IID Gaussian noise and give one dependent white-noise example.

Random walks and unpredictable increments

BEGINNER

ONE-SENTENCE MEANING

A random walk is a level process formed by accumulating innovations, so shocks persist in the level even when future increments are unpredictable.

MEMORY JOGGER

The level remembers every past shock, while the next change may have no directional memory. High persistence in prices does not by itself imply forecastable returns.

WHY IT MATTERS IN QUANT RESEARCH

Random walk behavior is a critical baseline for price models and trading claims. A strategy should outperform a properly specified no-predictability benchmark after costs, rather than merely fitting the obvious persistence of levels.

FORMULA INTUITION

P_t=P_(t-1)+epsilon_t or log P_t=log P_(t-1)+epsilon_t. Conditional expectation of the next increment is zero in the simplest form, while level variance grows with horizon.

SIMPLE MARKET EXAMPLE

Regressing tomorrow's price level on today's often gives an impressive fit because the level is persistent. Predicting tomorrow's return may still perform no better than zero.

PYTHON / SYSTEM IMPLEMENTATION

Benchmark level forecasts with the last observation and return forecasts with the historical or zero mean. Evaluate rolling forecast error and directional value without random row splits.

CONFUSIONS TO AVOID

Random walk does not require normal innovations or constant volatility, and real prices can include drift, jumps, and microstructure. A high level R-squared is not evidence of exploitable prediction.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast service includes naive random-walk baselines in every price experiment. Promotion requires incremental out-of-sample value and economically feasible execution relative to that baseline.

RELATED CONCEPTS AND QUICK RECALL

Related: unit root, martingale difference, naive forecast, persistence. Recall task: Explain why price levels are easy to predict numerically but returns remain difficult, and choose the correct baseline for each.

Drift is a small average change

BEGINNER

ONE-SENTENCE MEANING

Drift is a systematic expected increment in a process, often tiny relative to period-to-period noise and difficult to estimate precisely.

MEMORY JOGGER

A weak directional tendency can matter over long horizons while being nearly invisible on any single day. Estimation uncertainty and regime dependence dominate confident stories about short samples.

WHY IT MATTERS IN QUANT RESEARCH

Expected return, carry, and long-run price growth all resemble drift concepts. Quant decisions need to compare estimated drift with volatility, costs, and uncertainty rather than treating a positive sample mean as a stable edge.

FORMULA INTUITION

$X_t=X_{(t-1)}+\mu+\epsilon_t$ gives expected h-step change $h\mu$ under a stable additive model. Standard error of a mean shrinks with independent-equivalent observations, not simply row count.

SIMPLE MARKET EXAMPLE

A daily mean return of 0.04% alongside 1.5% daily volatility has a very low one-day signal-to-noise ratio. Several years of data may still leave a broad interval once dependence and regime change are considered.

PYTHON / SYSTEM IMPLEMENTATION

Estimate drift with units, horizon, sample span, effective observations, robust alternatives, and confidence or bootstrap intervals. Compare recent and long-run estimates and propagate uncertainty into sizing.

CONFUSIONS TO AVOID

Annualizing a noisy daily mean multiplies the point estimate but does not create evidence. Selecting the best historical window or excluding crises after inspection creates an upward-biased drift estimate.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric service publishes drift as an uncertain estimate object, and allocation consumes scenario draws rather than one plug-in number. Monitoring checks whether realized outcomes remain compatible with the approved range.

RELATED CONCEPTS AND QUICK RECALL

Related: sample mean, annualization, signal-to-noise ratio, expected return. Recall task: Assess whether a small positive daily mean is decision-relevant after costs and list the uncertainty information required.

Mean reversion needs a reference level

BEGINNER

ONE-SENTENCE MEANING

Mean reversion describes a tendency for deviations from a defined equilibrium or conditional mean to shrink over time.

MEMORY JOGGER

Reversion is not simply a price falling after it rose. The process needs a defensible reference, a speed, and evidence that the relationship survives changing regimes and trading costs.

WHY IT MATTERS IN QUANT RESEARCH

Spreads, valuation gaps, inventory imbalances, and short-horizon microstructure effects may revert. Estimating reversion informs entry, holding period, stop logic, and whether a level or difference should be modeled.

FORMULA INTUITION

An AR(1) deviation $x_t=\phi x_{t-1}+\epsilon_t$ reverts when $|\phi|<1$. Approximate half-life is $\log(0.5)/\log(|\phi|)$ for positive phi under a stable discrete-time model.

SIMPLE MARKET EXAMPLE

If a standardized spread moves from two to one to one-half with similar timing, it shows conditional decay toward zero. If the underlying hedge ratio changes, the apparent reversion may be a construction artifact.

PYTHON / SYSTEM IMPLEMENTATION

Fit the equilibrium and speed only on training data, test residual stationarity and parameter stability, and validate trading rules on later periods. Include costs, asynchronous prices, and structural-break scenarios.

CONFUSIONS TO AVOID

Negative return autocorrelation alone does not establish a tradable equilibrium. A broken relationship can look extremely far from its historical mean just before it diverges permanently.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The spread-construction service versions hedge parameters and equilibrium state. Strategy controls monitor deviation age, reversion speed, residual scale, and break alarms before allowing capital.

RELATED CONCEPTS AND QUICK RECALL

Related: AR(1), half-life, cointegration, spread. Recall task: Define the reference level and half-life for a candidate spread, then list evidence that would indicate a structural break instead.

Volatility clustering is memory in magnitude

BEGINNER

ONE-SENTENCE MEANING

Volatility clustering is the tendency for large absolute market moves to occur near other large moves and quiet moves near quiet moves.

MEMORY JOGGER

Direction may be hard to predict while scale remains persistent. A calm day says more about tomorrow's likely range than about tomorrow's sign.

WHY IT MATTERS IN QUANT RESEARCH

Risk limits, position sizing, option valuation, stop distances, and forecast intervals need time-varying scale. Constant-volatility assumptions underreact after shocks and overstate risk after prolonged calm.

FORMULA INTUITION

A simple EWMA variance is sigma_t^2=lambda sigma_(t-1)^2+(1-lambda)r_(t-1)^2. Clustering appears as positive autocorrelation in squared or absolute returns.

SIMPLE MARKET EXAMPLE

After a market shock, daily returns may alternate signs but remain large for weeks. Return ACF can stay near zero while squared-return ACF decays slowly.

PYTHON / SYSTEM IMPLEMENTATION

Plot rolling and EWMA volatility, squared-return ACF, and regime-conditioned distributions. Fit scale models on past data, archive forecasts before outcomes, and evaluate both coverage and interval width.

CONFUSIONS TO AVOID

Clustering is not proof that a particular volatility model is correct. A backward-looking estimator can lag abrupt state changes, and annualization by square-root time can fail under dependence and jumps.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk forecast service publishes point-in-time volatility and health status to sizing and limits. Calibration monitoring checks whether realized moves match promised distributions across regimes.

RELATED CONCEPTS AND QUICK RECALL

Related: conditional variance, EWMA, GARCH, realized volatility. Recall task: Explain how return direction can be unpredictable while risk is predictable, then design a basic coverage check.

Regime changes break pooled averages

BEGINNER

ONE-SENTENCE MEANING

A regime change is a material shift in the process governing location, scale, dependence, liquidity, or market behavior.

MEMORY JOGGER

One long history can contain several different data-generating environments. Pooling them produces an average that may describe no period well and can hide the transition risk between states.

WHY IT MATTERS IN QUANT RESEARCH

Strategy edges, correlations, costs, and volatility often change together during stress or structural reforms. Robust research tests whether decisions survive plausible states rather than optimizing one blended estimate.

FORMULA INTUITION

A regime model can write parameters theta_t=theta_(S_t), where latent or observed state S_t changes through time. Break detection concerns parameter stability, not merely a large individual observation.

SIMPLE MARKET EXAMPLE

A low-volatility relationship calibrated before a crisis can fail when correlations and spreads jump together. A full-sample estimate may look moderate even though neither calm nor crisis behavior is moderate.

PYTHON / SYSTEM IMPLEMENTATION

Use rolling diagnostics, change-point evidence, economic event labels, and predeclared state comparisons. Report sample support per regime and validate any state classifier without using future path information.

CONFUSIONS TO AVOID

Dividing history after seeing performance can manufacture favorable regimes, and every outlier is not a new state. Frequent adaptive refits can chase noise and forget rare but important stress observations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The monitoring layer detects parameter and data drift, while governance chooses fallback, retraining, or reduced limits. State definitions, detection latency, and response ownership are versioned system components.

RELATED CONCEPTS AND QUICK RECALL

Related: structural break, change point, local stationarity, stress test. Recall task: Describe how pooled correlation can misrepresent both calm and crisis states and propose a point-in-time response policy.

ACF estimates need uncertainty

BEGINNER

ONE-SENTENCE MEANING

A sample autocorrelation function is a noisy set of estimates across lags, so visible spikes require uncertainty, multiplicity, and dependence context.

MEMORY JOGGER

Every lag inspected is another chance for noise to look structured. A chart without pair counts, confidence guidance, or a prespecified lag range encourages pattern hunting.

WHY IT MATTERS IN QUANT RESEARCH

ACF plots guide baseline models, block lengths, and residual checks, but false spikes can lead to overfit lag structures. Market tails and changing variance make textbook bands only approximate.

FORMULA INTUITION

Under an ideal white-noise benchmark, rough pointwise bounds are plus or minus 1.96/sqrt(n). They are not simultaneous bands and can be wrong under heteroskedasticity or estimated preprocessing.

SIMPLE MARKET EXAMPLE

With 40 lags, about two may cross 95% pointwise bands by chance under a simple null. Selecting those two as a trading rule uses the same data twice.

PYTHON / SYSTEM IMPLEMENTATION

Fix maximum lag and transformations before inspection, report effective paired counts, and compare block-bootstrap or simulation envelopes. Confirm selected structure on later time windows.

CONFUSIONS TO AVOID

A spike just inside a band is not proof of no effect, and a spike outside is not proof of stable alpha. Overlapping returns and stale prices can create mechanical ACF patterns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The diagnostics service stores the complete lag family and null method. Model selection receives candidates from training diagnostics, while final evaluation occurs on untouched chronological data.

RELATED CONCEPTS AND QUICK RECALL

Related: confidence band, portmanteau test, multiple testing, bootstrap. Recall task: Estimate the false-spike problem across many lags and outline a confirmation procedure that does not reuse the same evidence.

Naive forecasts are hard baselines

BEGINNER

ONE-SENTENCE MEANING

A naive forecast uses a simple rule such as last value, historical mean, seasonal value, or zero return and establishes the minimum useful benchmark for a complex model.

MEMORY JOGGER

Complexity earns its place only by beating a cheap, reproducible alternative on the actual decision loss. Price levels are often difficult to beat with last-value forecasts even when model fit looks sophisticated.

WHY IT MATTERS IN QUANT RESEARCH

Baselines reveal whether a model learns beyond persistence, drift, seasonality, and volatility scale. They also provide safe fallbacks when data or model health degrades.

FORMULA INTUITION

Random-walk level forecast: $\hat{P}_{t+h|t}=P_t$. Mean-return forecast: $\hat{r}$=training mean; seasonal forecast uses the training average for the same calendar bucket.

SIMPLE MARKET EXAMPLE

A neural network can achieve high level R-squared yet have worse change forecast error than simply using today's price. The naive forecast captured persistence without pretending to predict returns.

PYTHON / SYSTEM IMPLEMENTATION

Implement immutable baseline models with the same features, cutoffs, horizons, costs, and scoring windows as challengers. Report incremental loss, uncertainty, and decisions rather than separate incomparable metrics.

CONFUSIONS TO AVOID

A weak baseline can make any model look useful, while a future-fitted seasonal average contaminates the comparison. Beating error metrics does not guarantee profitability after turnover and cost.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation platform runs approved baselines automatically and preserves one as the degraded-mode fallback. Promotion requires material, stable, and explainable improvement over the correct baseline.

RELATED CONCEPTS AND QUICK RECALL

Related: random walk, seasonal naive, benchmark, fallback. Recall task: Choose appropriate naive forecasts for a price level, return, and intraday volume curve, then define the comparison metric.

Time-ordered train, validation, and test

BEGINNER

ONE-SENTENCE MEANING

Time-series validation assigns earlier data to fitting and later data to selection and final evaluation so information flows in the same direction as deployment.

MEMORY JOGGER

The future must never teach the past. Validation guides choices and therefore becomes part of training history; a final test must remain untouched until the pipeline is frozen.

WHY IT MATTERS IN QUANT RESEARCH

Random splits leak regimes, neighboring observations, and overlapping outcomes across partitions. Chronological splits better estimate how a model handles drift, retraining, and genuinely unseen periods.

FORMULA INTUITION

A basic split satisfies $\max \text{time}(\text{train}) < \min \text{time}(\text{validation}) < \min \text{time}(\text{test})$, with gaps added for lookback, publication delay, and label overlap. Every preprocessing fit follows the same boundaries.

SIMPLE MARKET EXAMPLE

Train on 2015-2021, select on 2022-2023, and test once on 2024-2025 after freezing. If 2024-2025 results inspire changes, that interval is no longer an untouched test.

PYTHON / SYSTEM IMPLEMENTATION

Represent split boundaries as versioned artifacts and pass them into feature fitting, model fitting, and scoring. Hash the frozen pipeline before test access and log every reveal.

CONFUSIONS TO AVOID

Chronological order alone does not prevent leakage from full-sample scaling, revised data, or overlapping targets. One favorable test period also does not establish stability across multiple future regimes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment service controls dataset access and separates development from sealed evaluation. Model registry entries bind code, data snapshot, split, preprocessing, and metrics into one auditable result.

RELATED CONCEPTS AND QUICK RECALL

Related: holdout, walk-forward, embargo, model freeze. Recall task: Design a three-way split for a one-month target and explain when the test set becomes development data.

Walk-forward evaluation mimics operation

BEGINNER

ONE-SENTENCE MEANING

Walk-forward evaluation repeatedly fits on past data and scores the next chronological block, approximating how a model would be updated and used through time.

MEMORY JOGGER

Move the cutoff forward, refit what production would refit, and never revise earlier predictions. The procedure tests both the model and the retraining policy.

WHY IT MATTERS IN QUANT RESEARCH

Rolling and expanding walk-forward designs reveal regime sensitivity, warm-up behavior, and operational turnover. They produce a sequence of truly out-of-sample forecasts suitable for calibration and PnL analysis.

FORMULA INTUITION

At origin t_j, fit on window W_j and predict H_j with W_j entirely earlier than H_j. Aggregate losses only after preserving each forecast's origin, model version, and eligible outcomes.

SIMPLE MARKET EXAMPLE

A monthly-refit model trains through January and predicts February, then trains through February and predicts March. February outcomes cannot influence the forecasts already archived for February.

PYTHON / SYSTEM IMPLEMENTATION

Build an origin table containing train start, train end, prediction start, prediction end, purge gap, and artifact hash. Parallelize origins only if each job reads an immutable cutoff snapshot.

CONFUSIONS TO AVOID

Overlapping forecast blocks create dependent errors and need appropriate uncertainty. Reoptimizing the walk-forward schedule after seeing results is another hyperparameter search requiring nested evaluation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The backtest orchestrator produces an append-only forecast ledger identical in shape to live predictions. Monitoring and promotion consume that ledger rather than reconstructed in-sample fitted values.

RELATED CONCEPTS AND QUICK RECALL

Related: rolling origin, expanding window, refit cadence, prequential evaluation. Recall task: Specify a monthly walk-forward schedule and identify which objects must be refit and archived at every origin.

Lookahead leakage hides in preprocessing

BEGINNER

ONE-SENTENCE MEANING

Lookahead leakage occurs whenever a feature, transform, sample, or model uses information unavailable by the intended decision cutoff.

MEMORY JOGGER

Leakage is a timing violation, not merely a positive row shift. Means, scales, universes, revisions, labels, and data-cleaning thresholds can all carry future knowledge.

WHY IT MATTERS IN QUANT RESEARCH

Financial signals are weak enough that small leaks can dominate true edge and survive apparently rigorous backtests. Point-in-time discipline must cover the complete pipeline, not only the final model matrix.

FORMULA INTUITION

For prediction time t, every input artifact A must satisfy availability(A)<=t and must have been fit only from records satisfying their own legal cutoffs. Event date is not sufficient evidence.

SIMPLE MARKET EXAMPLE

Standardizing 2010 data with mean and variance calculated through 2025 makes the earlier feature depend on later crises. The leak remains even if the model itself trains only through 2010.

PYTHON / SYSTEM IMPLEMENTATION

Encapsulate imputation, clipping, scaling, selection, and estimation in fit/transform objects executed inside each split. Add future-mutation tests and lineage queries that identify the latest source timestamp per prediction.

CONFUSIONS TO AVOID

Shifting the final feature table backward does not repair upstream full-history fitting. Revised fundamentals and current index membership can leak future survival or correction information without any obvious future timestamp.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The point-in-time platform enforces cutoff-aware reads and stores cell-level or artifact-level lineage. Evaluation blocks datasets lacking availability time, vintage, or preprocessing fit boundaries.

RELATED CONCEPTS AND QUICK RECALL

Related: preprocessing leakage, survivorship bias, revision time, information set. Recall task: Audit one normalized fundamental feature from raw release through model input and list every timestamp that can cause leakage.

Overlapping horizons reuse outcomes

BEGINNER

ONE-SENTENCE MEANING

Overlapping forward returns share many of the same future price changes, creating strong dependence between neighboring labels and less information than the row count suggests.

MEMORY JOGGER

Draw the label intervals before counting examples. Two rows that reuse most outcome days are not two independent confirmations of a signal.

WHY IT MATTERS IN QUANT RESEARCH

Daily evaluation of monthly targets can exaggerate precision, smooth performance, and leak nearly identical outcomes across train and validation. Dependence-aware inference and purged splitting are required.

FORMULA INTUITION

An h-day log return $y_t=\sum_{j=1}^{h} r_{t+j}$ shares h-1 one-day returns with y_{t+1}. Even independent daily returns therefore induce label autocovariance.

SIMPLE MARKET EXAMPLE

Twenty-day labels created every day reuse nineteen days between neighbors. A random split can place nearly the same market path on both sides of the evaluation boundary.

PYTHON / SYSTEM IMPLEMENTATION

Store label_start and label_end timestamps, construct overlap groups, purge training rows whose outcome windows touch validation, and use block or cohort inference. Report nonoverlapping anchor results as a robustness view.

CONFUSIONS TO AVOID

More frequent labels may help specialized likelihood models but do not justify IID errors. An embargo chosen by row count can fail when calendars, missing sessions, or variable horizons alter elapsed overlap.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label service publishes interval metadata consumed by splitters, resamplers, and evidence reports. Promotion gates compare nominal labels, unique outcome windows, and effective sample support.

RELATED CONCEPTS AND QUICK RECALL

Related: purging, embargo, effective sample size, forward return. Recall task: Map the overlap between adjacent 20-day labels and design a split that prevents shared outcomes across partitions.

Backtest timestamps must model decisions

BEGINNER

ONE-SENTENCE MEANING

A valid backtest distinguishes observation, feature availability, decision, order submission, execution, and outcome timestamps for every simulated action.

MEMORY JOGGER

The signal clock and fill clock are not the same. A close-derived signal cannot trade at that same close unless an explicit auction or latency model makes the information and execution feasible.

WHY IT MATTERS IN QUANT RESEARCH

One-bar timing assumptions can turn a weak idea into a strong equity curve. Precise clocks connect time-series research to market mechanics, costs, and production latency.

FORMULA INTUITION

Require observation_time<=availability_time<=decision_time<=order_time<=fill_time. Outcome windows begin only after the executable position is established under the chosen convention.

SIMPLE MARKET EXAMPLE

A signal using the official close is finalized after the closing auction, so filling at that printed close is generally optimistic. A next-open or next-valid-quote fill better matches information flow.

PYTHON / SYSTEM IMPLEMENTATION

Persist a forecast and order ledger with all timestamps, source versions, latency, side, quantity, benchmark, and fill rule. Add assertions that no fill precedes the feature's latest availability.

CONFUSIONS TO AVOID

Bar labels such as 16:00 do not prove the close was known or tradable exactly then. Ignoring queue, spread, market impact, and rejected orders still creates implementation shortfall even with correct chronological ordering.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The backtester shares clock semantics with live execution and consumes point-in-time feature artifacts. Reconciliation compares simulated and live timestamp paths to expose latency or fill-model drift.

RELATED CONCEPTS AND QUICK RECALL

Related: decision time, execution latency, next-bar rule, implementation shortfall. Recall task: Draw the complete timestamp chain for a close-based signal and identify the earliest defensible market order fill.

Residual diagnostics test what remains

BEGINNER

ONE-SENTENCE MEANING

Residual diagnostics examine forecast errors for bias, serial structure, changing scale, tails, and state dependence that the fitted model failed to explain.

MEMORY JOGGER

A model is not finished when coefficients are estimated. Its errors should be compatible with the assumptions used for intervals, decisions, and claimed predictability.

WHY IT MATTERS IN QUANT RESEARCH

Autocorrelated residuals imply missed time structure, biased residual means imply calibration error, and clustered squared residuals imply changing uncertainty. These failures can damage position sizing even when average forecast loss improves.

FORMULA INTUITION

Residual e_t=y_t-y_hat_(t|t-1). Diagnostics include mean, ACF(e), ACF(e^2), conditional quantiles, coverage, and loss by regime, all computed from out-of-sample forecasts.

SIMPLE MARKET EXAMPLE

A return model can have uncorrelated errors but repeated interval misses during volatile periods. Directional specification may be adequate while its uncertainty model is unsafe.

PYTHON / SYSTEM IMPLEMENTATION

Archive predictions before outcomes, then generate residual panels by horizon, asset, calendar bucket, and market state. Apply fixed tests and visual checks with dependence-aware uncertainty.

CONFUSIONS TO AVOID

In-sample residuals are artificially favorable and do not substitute for forecast errors. Passing a portmanteau test does not prove independence, normality, calibration, or economic usefulness.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation service emits a standardized residual artifact used by model review and live monitoring. Alert thresholds connect specific error patterns to recalibration, fallback, or retraining actions.

RELATED CONCEPTS AND QUICK RECALL

Related: forecast error, calibration, portmanteau test, conditional variance. Recall task: Design a residual scorecard that separates directional misspecification from interval and volatility failure.

SYSTEM DESIGN

A time-series feature pipeline is stateful

BEGINNER

ONE-SENTENCE MEANING

A time-series feature pipeline transforms ordered point-in-time data through stateful operations whose fitted parameters and rolling history must be reproducible at every prediction cutoff.

MEMORY JOGGER

The model is more than the final estimator. Calendars, joins, lags, windows, imputers, scales, feature selectors, and warm-up state all shape what the model actually sees.

WHY IT MATTERS IN QUANT RESEARCH

Training-serving consistency is difficult when notebook code recomputes history differently from a streaming service. Treating preprocessing as versioned model state prevents silent divergence and makes backtests replayable.

FORMULA INTUITION

Pipeline output $F_t=\text{Transform}(A_{\le t}; \text{state}_{(t-1)}, \text{config})$, followed by $\text{state}_t=\text{Update}(\text{state}_{(t-1)}, A_t)$. Causal order and idempotent updates are explicit design properties.

SIMPLE MARKET EXAMPLE

A rolling z-score built in batch with a library default may include the current close, while the live service uses only prior closes. Identical formula names then produce different signals.

PYTHON / SYSTEM IMPLEMENTATION

Use typed transforms with fit, transform, update, snapshot, and restore interfaces. Add batch-versus-stream equivalence tests, cutoff mutation tests, schema validation, and deterministic warm-up rules.

CONFUSIONS TO AVOID

Recomputing from corrected history can change past features that production originally saw. Online updates without event deduplication can double-count late or retried messages.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature platform owns state snapshots, lineage, and replay, while the model registry references exact transform versions. Production can roll back features independently of strategy logic when a data defect appears.

RELATED CONCEPTS AND QUICK RECALL

Related: feature store, online state, idempotency, training-serving skew. Recall task: List the state that must be saved for a rolling normalized feature and design a batch-stream equivalence test.

SYSTEM DESIGN

Storage schemas preserve time semantics

BEGINNER

ONE-SENTENCE MEANING

A reliable time-series schema stores entity identity, event time, availability time, value, quality state, and version so historical knowledge can be reconstructed.

MEMORY JOGGER

One timestamp rarely carries enough meaning. Query speed is useful only after the schema preserves what happened, when it happened, when it became known, and whether it was later revised.

WHY IT MATTERS IN QUANT RESEARCH

Point-in-time backtests, deduplication, audit, and live replay all depend on temporal keys. Wide convenience tables without vintages can silently overwrite the evidence required to reproduce past decisions.

FORMULA INTUITION

A bitemporal record commonly has valid interval [valid_from,valid_to) and system interval [known_from,known_to). Queries select values valid for the market date and known by the research cutoff.

SIMPLE MARKET EXAMPLE

A company filing may be economically valid for a quarter, published weeks later, ingested later still, and revised after that. One quarter_end column cannot represent these different clocks.

PYTHON / SYSTEM IMPLEMENTATION

Use stable instrument IDs, UTC event timestamps, local session IDs, availability timestamps, revision IDs, quality flags, and immutable raw payload references. Partition for access patterns without dropping keys from the logical model.

CONFUSIONS TO AVOID

Using ticker as a permanent key breaks through renames and reuse. Upserts that overwrite revised values make historical feature replay impossible even if the current table looks clean.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research database exposes as-of query APIs and immutable snapshots to feature jobs. Catalog metadata binds schema, units, calendar, and lineage to every model artifact.

RELATED CONCEPTS AND QUICK RECALL

Related: bitemporal data, security master, revision, immutable snapshot. Recall task: Design the primary temporal fields for a revised fundamental observation and explain how to query what was known on a past date.

SYSTEM DESIGN

Monitoring detects data and process drift

BEGINNER

ONE-SENTENCE MEANING

Time-series monitoring tracks whether incoming data, temporal structure, forecasts, and operational behavior remain compatible with the approved research system.

MEMORY JOGGER

Monitor the clock, the values, the relationships, and the decision consequences. Stable PnL can temporarily hide stale feeds, rotating features, or undercovered risk forecasts.

WHY IT MATTERS IN QUANT RESEARCH

Coverage loss, latency, distribution shift, autocorrelation change, residual bias, and cost drift can each invalidate a model differently. Alerts need diagnostic context and preapproved actions rather than a single generic threshold.

FORMULA INTUITION

Health combines freshness, expected-row coverage, quality error rate, feature drift, forecast calibration, residual dependence, and decision sensitivity. Thresholds should reflect sampling uncertainty and materiality.

SIMPLE MARKET EXAMPLE

A quote feed begins updating slowly, creating repeated midpoints and artificial low volatility. Signal values remain numerically stable, but age, zero-return frequency, and realized forecast misses deteriorate.

PYTHON / SYSTEM IMPLEMENTATION

Archive reference distributions and point-in-time forecasts, then monitor by instrument, session, horizon, and regime. Define warning, degraded, and hard-stop levels with owners, fallbacks, and rollback artifacts.

CONFUSIONS TO AVOID

Automatic retraining can absorb bad data or temporary stress and make the failure harder to diagnose. Drift in a normalized feature may arise from either raw values or its moving denominator, which should be monitored separately.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane binds model health to capital limits and can switch to conservative baselines without waiting for a code deployment. Incident logs connect alerts, decisions, and eventual remediation.

RELATED CONCEPTS AND QUICK RECALL

Related: data drift, concept drift, observability, fallback. Recall task: Draft three alert tiers for stale data and forecast undercoverage, including the exact operational response at each tier.

SYSTEM DESIGN

Reproducibility requires temporal lineage

BEGINNER

ONE-SENTENCE MEANING

Temporal lineage records the exact data versions, cutoffs, transformations, state, code, and decisions needed to reproduce what a time-series system knew at a past moment.

MEMORY JOGGER

A result is reproducible only if the past information set can be replayed, not merely if today's code reruns. Mutable vendor history and revised calendars make this stricter than saving a notebook.

WHY IT MATTERS IN QUANT RESEARCH

Research review, incident analysis, model comparison, and regulatory audit all require evidence linking forecasts and trades to their source state. Without lineage, improvements cannot be separated from changed data.

FORMULA INTUITION

Artifact identity can hash data snapshot, schema, calendar, transform config, code commit, split, model state, and environment. The forecast ledger maps each prediction to that immutable identity.

SIMPLE MARKET EXAMPLE

A backtest rerun improves after a vendor corrects old prices. Without the original snapshot, the team cannot know whether the model changed or the historical inputs changed.

PYTHON / SYSTEM IMPLEMENTATION

Store immutable raw partitions or content hashes, dependency manifests, feature lineage, state checkpoints, and output ledgers. Add one-command replay tests for selected historical decisions and compare values bit-for-bit where practical.

CONFUSIONS TO AVOID

A random seed does not make mutable inputs reproducible, and a code commit does not capture database state. Exact numerical replay can still be insufficient if external calendars or service configurations were not versioned.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The artifact registry is the spine connecting data, feature, model, backtest, and deployment records. Promotion and incident tooling refuse untraceable outputs and surface the earliest changed dependency.

RELATED CONCEPTS AND QUICK RECALL

Related: artifact hash, data snapshot, lineage graph, replay. Recall task: List every dependency needed to reproduce one historical signal and explain how to distinguish a code change from a data revision.

ANNOTATED CHART

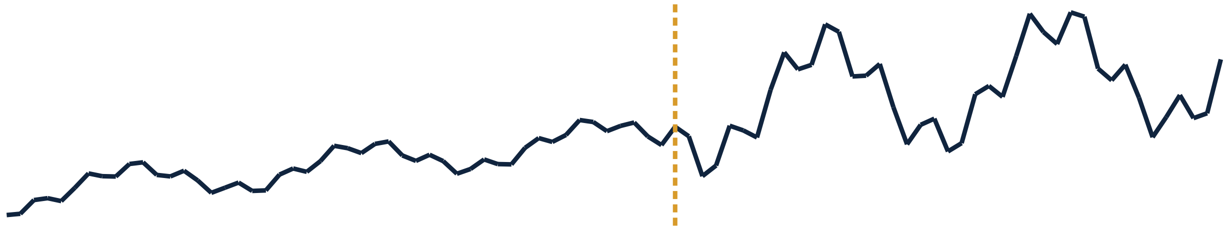
One price path, several useful views

BEGINNER

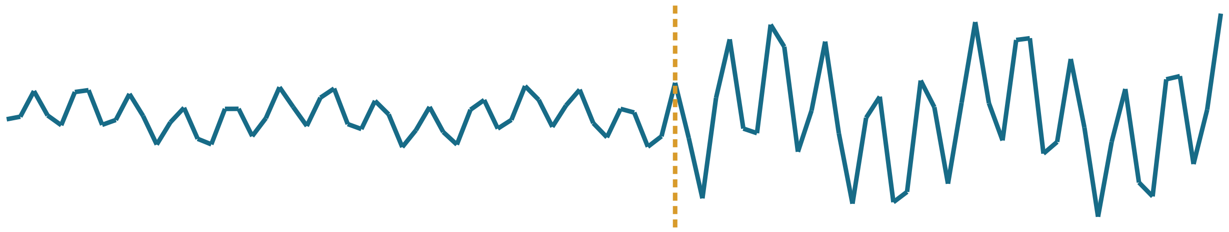
READING OBJECTIVE

The same synthetic market path is shown as level, one-period return, and rolling scale. Each representation answers a different question; none is a universally correct replacement for the others.

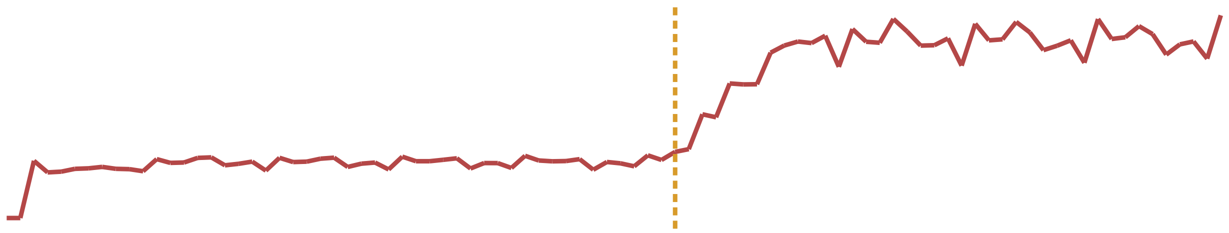
Price level



One-period return



Trailing 10-period volatility



INTERPRETATION

The level remains visually smooth because changes accumulate, while returns reveal sign and shock timing. Rolling volatility rises only after enough larger moves enter its lookback, illustrating estimation delay and why state variables must be lagged to the decision cutoff.

Return formulas and wealth interpretation

BEGINNER

DOLLAR CHANGE

Delta $P_t=P_t-P_{(t-1)}$. It preserves price units and suits tick-value or spread questions, but it cannot compare assets with different starting prices without further scaling.

SIMPLE RETURN

$r_t=P_t/P_{(t-1)}-1$. It maps directly to one-period wealth and cross-sectional portfolio aggregation when weights share the same interval.

LOG RETURN

$\ell_t=\log(P_t/P_{(t-1)})$. It adds through time and supports continuous compounding, but weighted asset log returns are not exact portfolio log return.

CUMULATIVE WEALTH

$W_t=W_0 \prod(1+r_i)$. It preserves sequence, costs, leverage, and drawdown; summing simple returns does not reproduce it.

WORKED PATH

Prices 100 -> 110 -> 99 give simple returns +10% and -10%. Wealth ends at 99, cumulative return is -1%, and log returns sum to log(0.99).

CHOICE RULE

Use simple returns for realized portfolio wealth, log returns for additive time decomposition, and dollar changes where economic exposure is genuinely in price units. Declare the convention in every metric.

VALIDATION RULE

Reconstruct the final price ratio from cumulative returns and compare raw versus adjusted inputs. Large disagreements usually reveal gaps, corporate actions, units, or alignment defects.

QUICK RECALL

Explain why +50% then -50% loses wealth, why log returns are additive, and why adjusted prices are analytical artifacts rather than executable historical quotes.

Autocorrelation patterns and traps

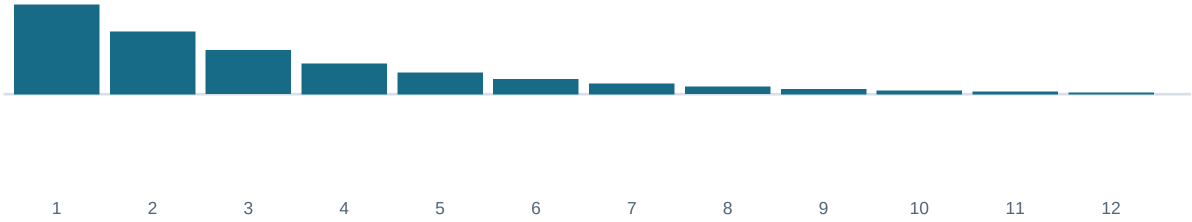
BEGINNER

THREE STYLIZED PROCESSES

Bars illustrate expected shapes rather than empirical guarantees. Use them to propose baseline models, then rely on chronological validation and residual diagnostics rather than pattern matching alone.

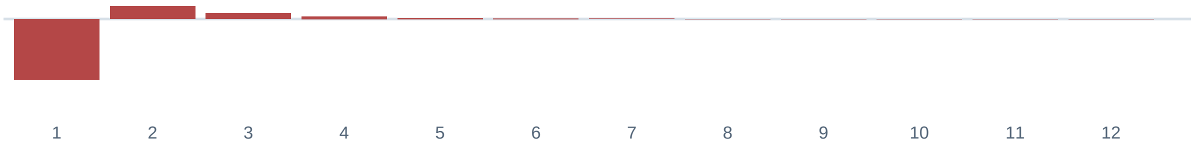
Positive persistence

ACF decays gradually; a direct lag-one mechanism can propagate to farther lags.



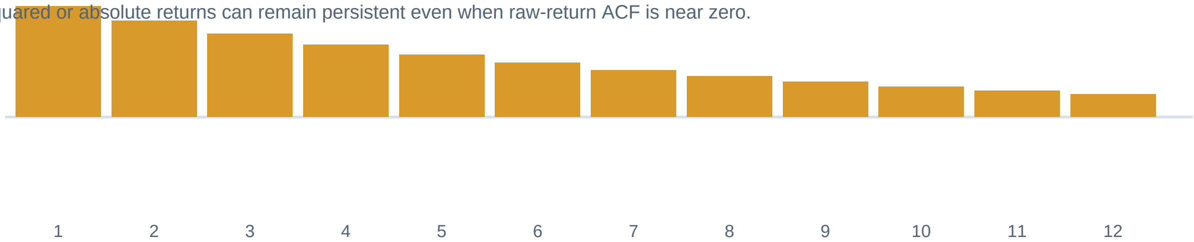
Short reversal

Negative lag one can reflect true reversal, bid-ask bounce, or stale-price correction.



Volatility memory

Squared or absolute returns can remain persistent even when raw-return ACF is near zero.



USE WITH DISCIPLINE

Fix the transformation and maximum lag before inspection, show pair counts and uncertainty, and confirm selected lags on later data. Mechanical overlap, smoothing, staleness, and microstructure can all create convincing but non-economic shapes.

DECISION TABLE

Choosing the observation grid

BEGINNER

DECISION PRINCIPLE

Select the grid that matches the information arrival, action cadence, and execution mechanism. Then test nearby grids to reveal microstructure artifacts and horizon-sensitive conclusions.

GRID	PRIMARY BENEFIT	MAIN RISK	GOOD FIT
Trades or quotes	Exact event path	Noise and volume	Execution, book dynamics
Clock bars	Equal elapsed time	Unequal activity	Portfolio and intraday state
Tick bars	Equal event counts	Variable duration	Activity-normalized studies
Volume bars	Roughly equal traded size	Threshold overshoot	Liquidity and impact
Daily session bars	Clean operational cadence	Overnight mixing	Medium-horizon signals
Weekly or monthly	Lower noise and turnover	Few observations	Slow factors and allocation

VALIDATION CHECKLIST

Compare return distribution, ACF, volatility, cost, coverage, duration, and strategy turnover across adjacent grids. Preserve partial-bar rules and never let a resampled close enter a decision before its interval is finalized.

SYSTEM CONNECTION

Create all approved grids from the same immutable event source and version every boundary rule. A model artifact references one grid ID, allowing production and research to replay identical bar assignments.

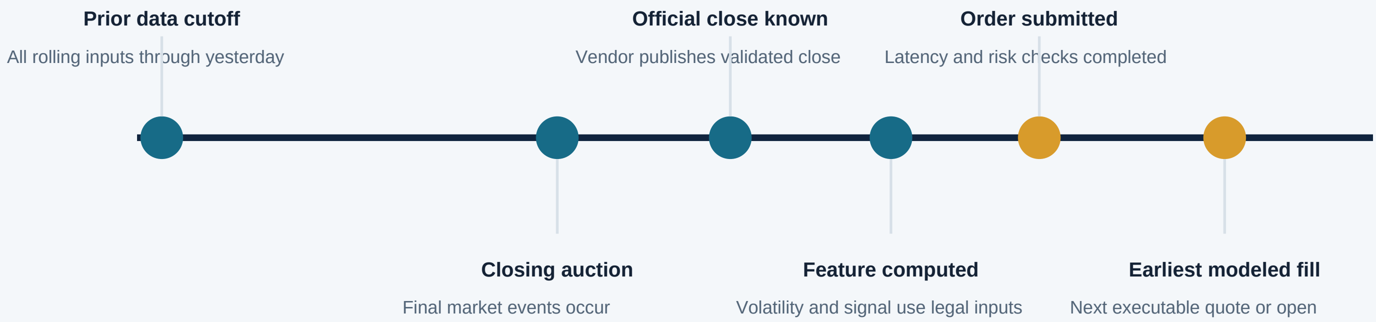
TIMELINE DIAGRAM

Point-in-time feature and trade timeline

BEGINNER

SCENARIO

A signal uses the official close and a trailing volatility estimate. The close must first be finalized and ingested; only then can the feature, order, and fill occur under a defensible latency model.



ILLEGAL SHORTCUT

Using the official close to compute the feature and simultaneously filling at that same close collapses observation, availability, decision, and execution into one timestamp. The backtest receives a price that was not yet actionable under its own information flow.

IMPLEMENTATION CONTRACT

Persist source event time, vendor availability, feature completion, decision, order, acknowledgment, and fill. Assert monotonic ordering and store the latest source availability used by every forecast.

ALTERNATIVE EXECUTION MODELS

A next-open fill is simple but exposes overnight risk; a next-quote fill needs intraday data and spread assumptions; an auction order requires a cutoff earlier than the auction. Select the model that matches the intended live process.

DIAGNOSTIC TABLE

Time-series failure modes and actions

BEGINNER

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Huge split-day return	Bad adjustment	Action ledger; raw price	Rebuild adjusted series
Many zero returns	Stale or filled data	Age and event count	Quarantine or model state
Strong lag-one reversal	Bid-ask bounce	Midquote versus last	Use valid benchmark
Excellent level R-squared	Random-walk persistence	Change forecast baseline	Score increments
Tiny standard errors	Overlapping labels	Interval overlap map	Purge and block inference
Live feature mismatch	State or boundary skew	Batch-stream replay	Unify transform artifact
Crisis interval misses	Volatility regime shift	Conditional coverage	Fallback and recalibrate
Backtest fills at close	Timing leakage	Timestamp inequality	Delay decision or fill

TRIAGE ORDER

Verify clock, identity, calendar, adjustment, and quality before changing a model. Then audit transformations and splits, then forecast assumptions, and only afterward tune complexity; a better model cannot repair an invalid series.

Formula sheet and interpretation map

BEGINNER

SIMPLE AND LOG RETURN

r_t=P_t/P_(t-1)-1; ell_t=log(1+r_t). Simple returns map to wealth, while log returns add exactly through adjacent time intervals.

CUMULATIVE WEALTH

W_t=W_0 product(1+r_i). Include fees, leverage, cash flows, and bankruptcy constraints in the same chronological update.

ROLLING MEAN

m_t=(1/L)sum_(i=0)^(L-1)X_(t-i). Lag the window when X_t is not finalized before the decision.

EWMA STATE

s_t=alpha X_t+(1-alpha)s_(t-1). Alpha controls reaction speed and effective memory; initialization and update order are part of the artifact.

AUTOCORRELATION

rho_k=Cov(X_t,X_(t-k))/Var(X_t). Report transformation, horizon, sample pairs, uncertainty, and state stability.

AR(1) BASELINE

X_t=c+phi X_(t-1)+epsilon_t. Values of phi near one imply persistence; |phi|<1 supports reversion under a stable model.

VOLATILITY SCALING

Under IID increments, sigma_h approximately sqrt(h)sigma_1. Treat this as a benchmark and test dependence, jumps, and calendar-time assumptions.

FORECAST ERROR

e_t=y_t-y_hat_(t|t-1). Diagnose mean, ACF, squared ACF, tails, and conditional interval coverage from archived out-of-sample predictions.

Worked example: prices to legal forecast

BEGINNER

1. VALIDATE THE INPUT

Suppose five adjusted closes are 100, 102, 101, 104, and 103. Confirm one asset, consecutive eligible sessions, one timezone, no split, and values known by each close cutoff.

2. COMPUTE SIMPLE RETURNS

The four returns are 2.000%, -0.980%, 2.970%, and -0.962%. Their arithmetic mean is about 0.757%, but that average does not equal compounded growth.

3. RECONCILE LOG RETURNS

Log returns are approximately 1.980%, -0.985%, 2.927%, and -0.966%. Their sum is log(103/100), so exp(sum)-1 equals the 3.000% cumulative simple return.

4. BUILD A LAGGED FEATURE

At the fifth close, a two-return trailing mean uses returns from 101 to 104 and 104 to 103 only if the decision occurs after that close is available. For a pre-close decision, lag one more period.

5. DEFINE THE LABEL

A next-session return label for the fourth close uses the fifth close and must not enter any fourth-close preprocessing. Label start and end times are stored for purging.

6. CHOOSE A BASELINE

For next price level, use the current price as a random-walk forecast. For next return, compare the model with zero and a training-only historical mean.

7. ADD EXECUTION TIMING

A signal finalized from the official close cannot generally fill at that same close. Use a next-open or next-valid-quote model with spread, latency, and overnight exposure.

8. PRESERVE THE AUDIT TRAIL

Save raw and adjusted price IDs, return convention, feature window, decision cutoff, label interval, baseline, forecast, order rule, and final metric. The number alone cannot reproduce the result.

REVIEW AND INDEX

Final active recall and system index

BEGINNER

SERIES CONTRACT

Define entity, event and availability times, timezone, session, frequency, grain, units, source, quality state, revision, and adjustment convention.

REPRESENTATION

Distinguish levels, dollar changes, simple returns, log returns, total returns, wealth, midquotes, and executable sides without mixing their economic meanings.

CONSTRUCTION

Audit OHLCV boundaries, calendars, stale observations, corporate actions, event-time bars, resampling, and backward as-of joins.

CAUSAL TRANSFORMS

Draw the timestamps for differences, lags, rolling windows, expanding state, smoothing, normalization, and forward labels.

BEHAVIOR

Interpret trend, seasonality, ACF, PACF, random walks, drift, mean reversion, volatility clustering, and regimes as conditional modeling claims.

VALIDATION

Use hard baselines, chronological splits, walk-forward origins, purging, untouched tests, residual diagnostics, cost models, and uncertainty.

EXECUTION

Enforce observation <= availability <= decision <= order <= fill and preserve side, latency, spread, quantity, and benchmark.

SYSTEMS

Version feature state, bitemporal storage, snapshots, artifact hashes, forecast ledgers, monitoring thresholds, fallback, and rollback.

FINAL RECALL

Explain how one timing error, one stale-price policy, and one overlapping label design can each create false predictability in an otherwise correct model.